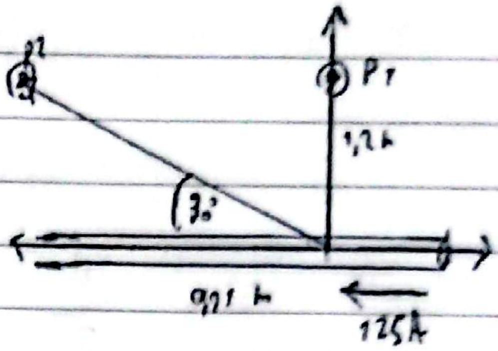
	práctica 6 - Fuentes de campo magnético	HOJA Nº:
	Física II	FECHA:

1)-

I = 125

$l = 1\,cm = 0,01\,m$

$r = 1,2\,m$



a)-

$$B = \frac{\mu_0}{4\pi}\cdot\frac{I\cdot l\cdot \sin\theta}{r^2} = \frac{4\pi\cdot 10^{-7}}{4\pi}\cdot\frac{125A\cdot 0,01m\cdot \sin(90°)}{(1,2m)^2}$$

$$\boxed{B = 8,68\cdot 10^{-8}\,T}$$

b)-

$$B = \frac{4\pi\cdot 10^{-7}}{4\pi}\cdot\frac{125A\cdot 0,01m\cdot \sin(30°)}{(1,2m)^2}$$

$$\boxed{B = 4,34\cdot 10^{-8}\,T}$$

2)-

$I = 1\,A$

$B = 0,5\cdot 10^{-4}\,T$

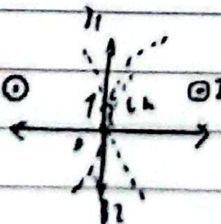
$$B = \frac{\mu_0\cdot I}{2\pi r} \Rightarrow 2\pi r = \frac{\mu_0 I}{B} \Rightarrow r = \frac{\mu_0 I}{B\cdot 2\pi}$$

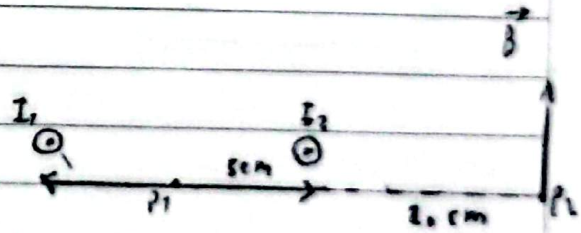
$$r = \frac{4\pi\cdot 10^{-7}\cdot 1A}{0,5\cdot 10^{-4}T\cdot 2\pi} = 4\cdot 10^{-3}\,m = \boxed{4\,mm}$$

3)-

$r = 10\,cm = 0,1\,m$

$I_1 = I_2 = 4A$





a)-

$$B = \frac{4\pi\cdot 10^{-7}\cdot 4A}{2\pi\cdot 0,05m} = 1,6\cdot 10^{-5}\,T$$

$$\boxed{B_1 + B_2 = 0}$$

b)-

$$B_1 = \frac{4\pi\cdot 10^{-7}\cdot 4A}{2\pi\cdot 0,3\,m} = 2,67\cdot 10^{-6}\,T$$

$$B_2 = \frac{4\pi\cdot 10^{-7}\cdot 4A}{2\pi\cdot 0,2\,m} = 4\cdot 10^{-6}\,T$$

$$\boxed{B_1 + B_2 = 6,67\cdot 10^{-6}\,T}$$

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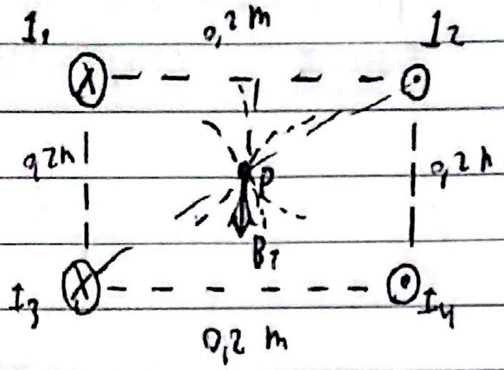
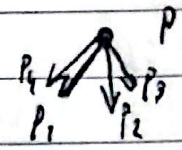
9-

$I_1 = I_2 = I_3 = I_4 = 5A$

$r = 0,2\,m$ de lado





$2r = \sqrt{(0,2m)^2 + (0,2m)^2} = \frac{\sqrt{2}}{5}$

$r = \frac{\sqrt{2}}{10} = 0,1414\,m$

$B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \cdot 10^{-7} \cdot 5A}{2\pi \cdot 0,1414\,m} = 7,07 \cdot 10^{-6}\,T$

$B_{total} =$

$B_T = 4 \cdot B \cdot \operatorname{sen}(45°) = 4 \cdot 7,07 \cdot 10^{-6}\,T \cdot \operatorname{sen}(45°) = 20 \cdot 10^{-6}\,T = \boxed{20\mu T}$

$\operatorname{cos}45$

5-

$I_1 = 25\,A$

$I_2 = 75A$

$r = 40\,cm = 0,4\,m$

$\frac{\mu_0 I_1}{2\pi r_1} = \frac{\mu_0 I_2}{2\pi r_2} \Rightarrow \frac{I_1}{r_1} = \frac{I_2}{r_2} = \frac{25A}{r_1} = \frac{75}{r_2}$

$\Rightarrow r_2 = 3r_1$

a- I_1 ⊙ —— desde I_1 $0,4 - x$ —— ⊙ I_2 $\qquad 0,4 - x = 3x \Rightarrow 4x = 0,4 \Rightarrow \boxed{x = 0,1\ m}$

b- I_1 ⊗ —— ⊙ I_2 $\qquad 0,4 + x$ $\qquad 0,4 + x = 3x \Rightarrow 2x = 0,4 \Rightarrow \boxed{x = 0,2\,m}$

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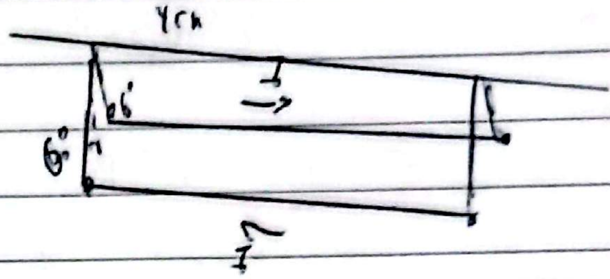
7)-



$l = 0{,}04\ m$

$m = 0{,}0125\ kg/m$

$\theta = 6°$

$d = 2 \cdot l \cdot \operatorname{sen}\theta = 2 \cdot 0{,}04\ m \cdot \operatorname{sen} 6° = 8{,}36 \cdot 10^{-3}\ m$

Eje y: $T \cdot \cos(6°) = \frac{W}{L} = \lambda \cdot g$

$\tan(6°) = \frac{F_B/L}{\lambda \cdot g} \Rightarrow \frac{F_B}{L} = \lambda \cdot g \cdot \tan(6°)$

Eje x: $T \cdot \operatorname{sen}(6°) = \frac{F_B}{L}$

$\frac{\mu_0 I^2}{2\pi d} = \lambda \cdot g \cdot \tan(6°)$ $\qquad 2 \cdot 10^{-7} \cdot \frac{I^2}{d} = \lambda \cdot g \cdot \tan(6°)$

$I^2 = \frac{d \cdot \lambda \cdot g \cdot \tan(6°)}{2 \cdot 10^{-7}} \Rightarrow I^2 \cong 538{,}36\ A^2$

$I = \sqrt{538{,}36} \cong \boxed{23{,}2\ A}$

8)-

a)

$r = 0{,}05\ m$

$N = 12$

$I = 4\ A$

$B = \frac{\mu_0 \cdot N I R^2}{2(x^2 + R^2)^{3/2}}$ $\qquad B = \frac{4\pi \cdot 10^{-7} \cdot 12 \cdot 4A\ (0{,}05\ m)^2}{2((0{,}05\ m)^2 + (0{,}15\ m)^2)^{3/2}}$

$\boxed{B \cong 1{,}71 \cdot 10^{-5}\ T}$

b)-

$B = \frac{\mu_0 \cdot N I R^2}{2 x^3} = \frac{4\pi \cdot 10^{-7} \cdot 12 \cdot 4A \cdot (0{,}05\ m)^2}{2 \cdot 3^3} = \boxed{2{,}79 \cdot 10^{-9}\ T}$

$\approx (r^2 + x^2)$

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11)-

$N = 100$
$r = 0{,}6\,m$
$I = 5A$

a)- $B_x = \dfrac{4\pi \cdot 10^{-7} \cdot 100 \cdot 5A \cdot (0{,}6m)^2}{2\left[(0{,}2m)^2 + (0{,}6m)^2\right]^{3/2}}$

$\boxed{B_x = 1{,}1 \cdot 10^{-4}\,T}$

b)- $\dfrac{1}{(x^2 + R^2)^{3/2}} = \dfrac{1}{8}\dfrac{1}{(0^2 + R^2)^{3/2}} \ldots \quad X = \pm\sqrt{3}R = \pm \boxed{1{,}04\,m}$

12)-

$N = 1$
$I = 5A$
$B = 10\,\mu T = 10 \cdot 10^{-6}\,T$

$B = \dfrac{\mu_0 N \cdot I}{2R} \Rightarrow 2R = \dfrac{\mu_0 N \cdot I}{B} \Rightarrow R = \dfrac{\mu_0 I}{2 \cdot B}$

$R = \dfrac{4\pi \cdot 10^{-7} \cdot 5A}{2 \cdot 10 \cdot 10^{-6}} = \dfrac{\pi}{10} \approx \boxed{0{,}31\,m}$

13)-

$L = 30\,cm = 0{,}3\,m$
$B = 5 \cdot 10^{-4}\,T$
$I = 1A$

$B = \dfrac{\mu_0 N I}{L} \Rightarrow B \cdot L = \mu_0 \cdot N \cdot I \Rightarrow N = \dfrac{B \cdot L}{\mu_0 \cdot I}$

$N = \dfrac{5 \cdot 10^{-4}\,T \cdot 0{,}3m}{4\pi \cdot 10^{-7} \cdot 1A}$ $\boxed{N \approx 119 \text{ voltas}}$

14)-

$N = 1000$
$L = 0{,}4\,m$
$B = 1 \cdot 10^{-4}\,T$

$B = \dfrac{\mu_0 \cdot N \cdot I}{L} \Rightarrow B \cdot L = \mu_0 \cdot N \cdot I \Rightarrow I = \dfrac{B \cdot L}{\mu_0 \cdot N}$

$I = \dfrac{1 \cdot 10^{-4}\,T \cdot 0{,}4\,m}{4\pi \cdot 10^{-7} \cdot 1000} \Rightarrow I = 0{,}03183\,A = \boxed{31{,}8\,mA}$

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FECHA:

15)-

Toroide

$r = 12\,cm = 0{,}12\,m$ $\quad r = 14\,cm = 0{,}14\,m$

$R = 15\,cm = 0{,}15\,m$

$I = 1{,}5\,A$

$B = 3{,}75\,mT = 3{,}75 \cdot 10^{-3}\,T$

$$B = \frac{\mu_0 N I}{2\pi r} \Rightarrow \frac{B \cdot 2\pi r}{\mu_0 \cdot I} = N$$

$$N = \frac{3{,}75 \cdot 10^{-3}\,T \cdot 2\pi \cdot 0{,}14\,m}{4\pi \cdot 10^{-7} \cdot 1{,}5\,A} = \boxed{1750 \text{ vueltas}}$$

16)-

$r_1 = 15\,cm = 0{,}15\,m$

$r_2 = 18\,cm = 0{,}18\,m$

$N = 250$

$I = 8{,}5\,A$

$$B = \frac{\mu_0 \cdot N \cdot I}{2\pi \cdot r} \quad (0{,}15\,m \leq r \leq 0{,}18\,m)$$

a)- $0{,}12\,m \Rightarrow \boxed{B = 0}$

b)- $0{,}16\,m$ $\quad B = \frac{4\pi \cdot 10^{-7} \cdot 250 \cdot 8{,}5\,A}{2\pi \cdot (0{,}16\,m)} = \boxed{2{,}66 \cdot 10^{-3}\,T}$

c)- $0{,}2\,m \Rightarrow \boxed{B = 0}$

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